

Declaration of Generative AI Tool Usage

in seminar papers, term papers, bachelor theses, and master theses

Title of the paper/thesis When Blocks Build Societies:
Multi-Agent Systems in Minecraft

Date of submission 2025-07-29

1. Overview on generative AI-based tools used for this paper or thesis

Indicate which tools that use generative AI you used. If you tick "Yes" on any of the tools, fill out the entire template. Otherwise, skip sections 2 and 3.

Tools	Usage	
	Yes	No
a. ChatGPT	☐	☒
b. DeepL Write	☐	☒
c. Microsoft 365 Copilot	☐	☒
d. Neuroflash	☐	☒
e. Grammarly	☐	☒
f. DeepL Translator	☐	☒
g. Google Translate	☐	☒
h. Perplexity	☒	☐
i. Elicit	☐	☒
j. Explainpaper	☐	☒
k. ResearchRabbit	☐	☒
l. GitHub Copilot	☒	☐
m. Other tool	☐	☒
n. Other tool	☐	☒
o. Other tool	☐	☒

2. Responsible use of the tools

Confirm the following statements.

Statements on responsible tool use	Confirmation: Yes
I/we are informed about the capabilities and limitations of the tools I/we used.	☒
I/we verified that the results given by the tools are accurate.	☒
I/we acknowledge that the responsibility for the paper/thesis lies with the author(s), not the tools or anybody else.	☒

3. Detailed activities for which the tools were used

Indicate which tool you used and to what extent for which activity.

Activity	Description	Tools used (if any) ¹	Description of the tool usage (type of usage, affected sections of the paper/thesis, etc.)
Ideation & conceptualization	- Generating ideas, re-search goals, aims, and questions - Identifying and defining relevant concepts	h, l	Perplexity (h) was used to brainstorm research questions, clarify concepts in multi-agent systems, and identify relevant theoretical frameworks. GitHub Copilot (l) was used for refining and structuring research aims and questions within the introduction and methodology sections.
Literature search and analysis	- Searching for relevant literature - Reviewing potentially relevant literature - Summaries of relevant literature	h	Perplexity (h) was used to identify, summarize, and synthesize relevant academic papers on multi-agent systems in Minecraft. It assisted in screening literature, extracting key findings, and generating summaries for the background and related work sections.
Methodology	- Searching for an appropriate methodology - Designing and tailoring the methodology to the	h, l	Perplexity (h) was used to explore and compare methodological approaches in MAS research. GitHub Copilot (l) assisted in drafting and structuring the methodology section, including the documentation approach for AI usage.
Coding	- Creating and documenting code, algorithms, software - Testing and debugging of existing code, algorithms, software - Understanding existing code, algorithms, software	l	GitHub Copilot (l) was used for generating, documenting, and reviewing code snippets and algorithms related to data analysis and figure/table generation. It also assisted in LaTeX formatting and troubleshooting.
Data collection and analysis	- Collection of primary or secondary data - Qualitative data analysis (including summarizing and coding) - Quantitative data analysis (including statistics) - Mathematical, computational, or other formal techniques for modeling, simulation, and analytics	l	GitHub Copilot (l) was used for generating and refining code for data analysis, table formatting, and figure integration in LaTeX. No primary data collection was performed; analysis focused on literature and case studies.
Interpretation and Validation	- Interpretation of results - Derivation of implications for research and practice - Verification of the overall replication/ reproducibility of results and other research outputs	h, l	Perplexity (h) was used to discuss and validate interpretations of findings from the literature. GitHub Copilot (l) assisted in drafting and refining the discussion and conclusion sections, ensuring logical consistency and clarity.

¹ Provide letters from the tool list in section 1.

Structuring and planning the text	– Outlining the paper/thesis – Outlining sections of the paper/thesis (e.g., bullet point lists per section)	l	GitHub Copilot (l) was used to generate outlines for the paper, suggest section structures, and create bullet-point lists for planning content flow.
Generating the text	– Generating text on various topics in different sections of the paper (including title and abstract)	h, l	Perplexity (h) was used to generate initial drafts and summaries for several sections (introduction, background, methodology, discussion, conclusion). GitHub Copilot (l) was used for drafting, expanding, and refining text throughout the manuscript, including the abstract and section transitions.
Translating text	– Translating text written by the authors – Translating text written by others		No AI tools were used for translation. All text was written and edited in English.
Reviewing & editing the text	– Critical review, feedback or revision on content, organization, or grammar of the paper – Proofreading – Rephrasing or paraphrasing text – Shortening / extending text	l	GitHub Copilot (l) was used extensively for reviewing, editing, rephrasing, and proofreading the text. It provided suggestions for improving clarity, grammar, and conciseness throughout the manuscript.
Presentation	– Structuring a presentation on the paper – Filling a presentation with content on the paper	h	Perplexity (h) was used to generate the initial outline and draft content for a 10-minute presentation based on the paper draft. The tool summarized key findings, structured the flow, and suggested slide content. All AI-generated material was reviewed and adapted by the author for clarity, accuracy, and suitability for the intended audience.
Citation Management	– Creation of reference list – Formatting of references	l	GitHub Copilot (l) assisted in formatting references and integrating them into the LaTeX document. Reference management was otherwise manual using BibLaTeX.
Further activities			None.

Is there anything else to declare regarding the use of AI tools that would highlight or limit your independent, definable performance in crafting this paper/thesis?

A unique aspect of this work is the transparent and systematic documentation of AI tool usage. The approach evolved from initially archiving full AI conversation logs to maintaining detailed, context-rich commit messages in the git version control system. Each significant change is accompanied by a descriptive commit message specifying the AI prompt, the rationale for accepting or modifying the output, and key learnings. This enables full traceability of AI influence and human oversight, ensuring transparency and reproducibility. All AI-generated content was critically reviewed, verified, and edited by the author to ensure academic integrity and originality.

4. Signature(s) of author(s)

	Author last name, first name(s)	Matriculation number	Date	Signature
1	Kiel, Jan-Philipp	7419035	2025-07-29	

Background on this declaration: Assessments at the university must allow students' performance to be determined. For this purpose, the student's own performance, which is contained in a paper/thesis, must be distinguished from the performance of others. It is also necessary to understand the tools used. As long as generative AI is relatively new in higher education, its use must be declared in detail. This serves the evaluation of the performance and assures students that their work will not be reassessed afterwards against the background of other norms.

Sources: Gimpel, H., Dilger, P. (2023). Declaration of Generative AI Tool Usage in seminar papers, term papers, bachelor theses, and master theses. University of Hohenheim, May 3, 2023.